

Little's law

$$E[N] = \lambda \times E[T]$$

(T is the time spent in a queue at stationarity)

For M/M/1/ ∞ ,

$$E[T] = \frac{1}{\lambda} \frac{\rho}{1-\rho} = \frac{1}{\mu - \lambda}$$

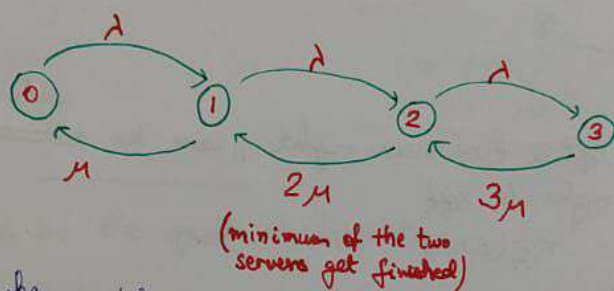
Find its distribution (ends with ...)

M/M/ ∞ / ∞

$\downarrow$ $\text{ref}(\lambda)$ $\downarrow$ $\text{exp}(\mu)$

S_1, S_2
 $\downarrow \quad \downarrow$
 X_1, X_2

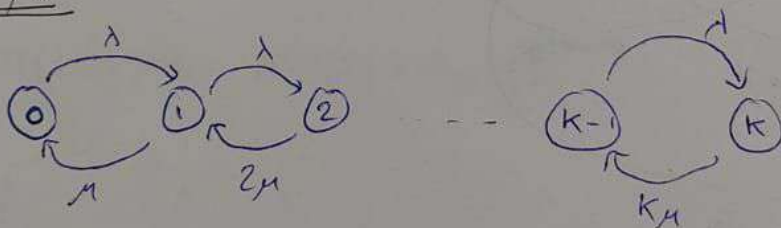
$\min(X_1, X_2) \rightarrow$ when you take minimum of two exp. r.v., you get r.v. which is sum of both



(Think about its discrete version using Bernoulli and geometric)

M/M/1/1
Geom/Geom/1/1

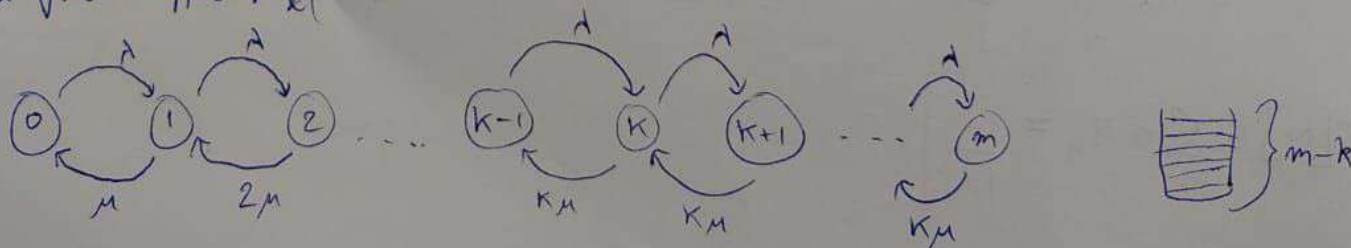
M/M/K/K



M/M/K/m

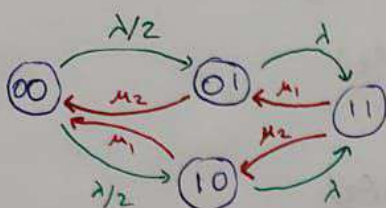
$m > k$

so a finite buffer of $(m-k)$



M/M/2/2

μ_1 μ_2



0 0 → empty
1 0 → 1st server busy
0 1 → 2nd server busy
1 1 → both busy

$$N(t) = \begin{cases} 0 \\ 1 \\ 2 \end{cases}$$

↓
tough to
use this
descriptor

→ instead of 11 you can also call full or f
and so on, just a representation

μ higher, so faster

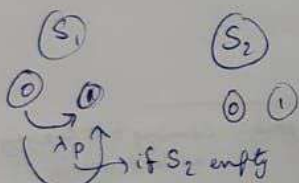
→ need to have an assignment policy to decide which
server to go to

→ assigning equally so $\lambda/2$

→ instead of $\lambda/2$, it can
be $\lambda(p)$ and $\lambda(1-p)$

→ find stationary distribution

S_1 and S_2 can't be decoupled
directly:



$\lambda \cdot 1 \rightarrow$ if S_1 full so has to require some info S_2 , there is a correlation

Finding cost of running the server:

$$\begin{aligned} C(e) &= c_1 \\ C(s_1) &= c_2 \\ C(s_2) &= c_3 \\ C(s_3) &= c_4 \end{aligned}$$

$$\pi_1 \times c_1 + \pi_2 \times c_2 + \dots$$

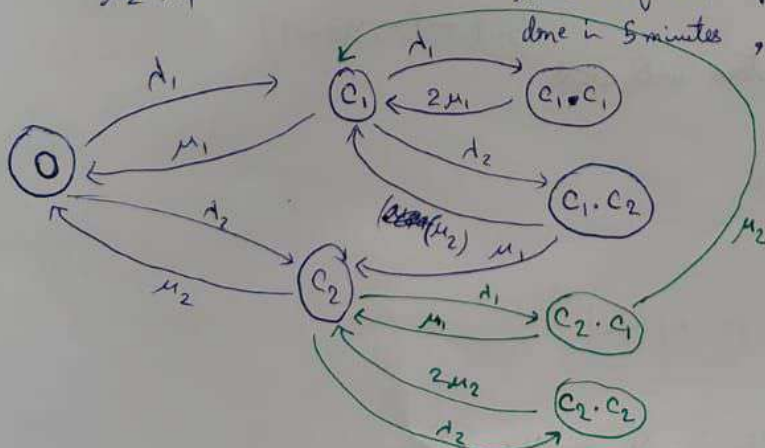
Rich / poor different requirements

rich more work
 $\lambda_1 \exp(\mu_1) \rightarrow c_1$ (class 1)

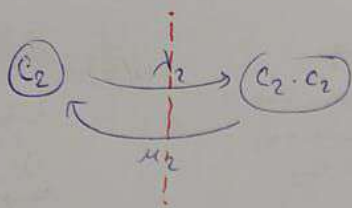
$\lambda_2 \exp(\mu_2) \rightarrow c_2$ (class 2)
 $\mu_2 > \mu_1$

① ①

Servers serve with constant rates, deterministic now, not
random anymore, if you bring in 5 minutes of work will be
done in 5 minutes, servers are identical



→ $C_1 \cdot C_2$ and $C_2 \cdot C_1$ can be
merged but that won't give
the details of which came first



$$\lambda_2 \pi(c_2) = \mu_2 \pi(c_2, c_2)$$

$$\frac{\lambda_2}{\mu_2} \pi(c_2) = \pi(c_2, c_2)$$

$$\pi(c_1) \frac{\lambda_1}{\mu_1} = \pi(c_1, c_1)$$

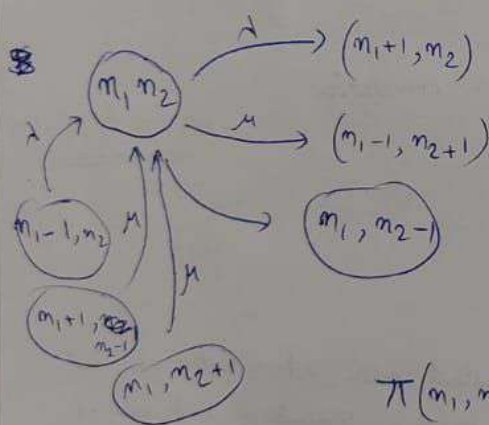
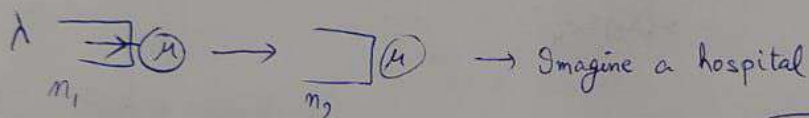
$$\frac{\lambda_1}{\mu_1} \pi(c_2) = \pi(c_1, c_2)$$

↓
When c_1, c_2 and c_2, c_1 are merged as otherwise there are odd rows

$$\pi(c_1, c_2) (\mu_2 + \mu_1) = \lambda_2 \pi(c_1) + \lambda_1 \pi(c_2)$$

→ Detail balance is summed term by term

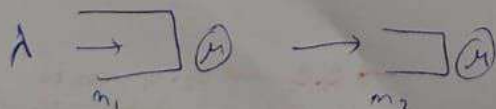
Tandem Queue



Blocking prob: coming but leaving without service
in M/M/k/k
it is simply π_k

$$\pi(n_1, n_2) (\lambda + 2\mu) = \lambda \pi(n_1 - 1, n_2) + \mu (\pi(n_1 + 1, n_2 - 1) + \pi(n_1, n_2 + 1))$$

now, equate which term with which



Treat like M/M/1 queue

$$\pi(n_1) = (1 - \rho_1) \rho_1^{n_1}$$

$$\pi(n_2) = (1 - \rho_2) \rho_2^{n_2}$$

$$\pi(n_1, n_2) = (1 - \rho_1) \rho_1^{n_1} (1 - \rho_2) \rho_2^{n_2} \rightarrow \text{Derive this}$$

⇒ Rate at which you enter a state i is equal to rate at which you leave state i .

Subsets of state

$$M \subseteq S$$

$$\pi_i \sum_{\substack{j \neq i \\ j \in M}} q_{ij} = \sum_{\substack{j \neq i \\ j \in M}} \pi_j q_{ji} \rightarrow \text{Partial Balance eq}^n$$

if holds
for all i and all subsets

Global Balance Equation

$$\sum_{j \neq i} \pi_j q_{ji} = \pi_i \sum_{j \neq i} q_{ij} \Rightarrow \pi Q = 0$$

check for all $i \rightarrow 1$ to M
which equals to $\pi Q = 0$

$$\pi_i q_{ik} = \pi_k q_{ki} \rightarrow \text{Detailed Balance eq}^n$$

$$\pi_i q_{i1} + \pi_i q_{i2} + \dots + \pi_i q_{im} = \pi_1 q_{1i} + \pi_2 q_{2i} + \pi_3 q_{3i} + \dots + \pi_m q_{mi}$$

→ see if it's a valid solution and check for all i , to see if global balance eqⁿ holds

→ for Global we need ^{to check} eqⁿ for all i

$M/M/1/1$

Arrival process
 $\rho(\lambda)$

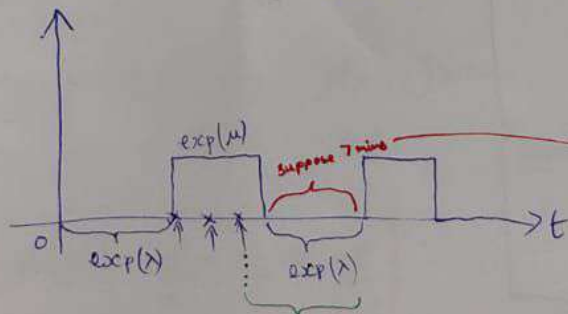
Service time

No. of jobs in the system at a time

No. of servers

$$\{x(t), t \geq 0\} \quad \bullet \quad x(t) = \begin{cases} 1 & \text{if server is busy} \\ 0 & \text{if server is empty} \end{cases}$$

$\mu \rightarrow$ the server samples according to exp. distribution the time taken to do the same # units of work.



exp starts here (the last arrival)
but memory less property so
taken when server became idle (think!)

• arrivals when server is busy are lost

$$P(x(t+\tau)=1/x(t)=0)$$

• It doesn't matter when the previous guy arrived, when you have to calculate after time t , the previous info. doesn't matter

$$\lambda \pi_0 = \mu \pi_1$$

$$\pi_1 = \left(\frac{\lambda}{\mu}\right) \pi_0 = \rho \pi_0$$

ρ : utilisation factor

$$\pi_2 = \rho^2 \pi_0$$

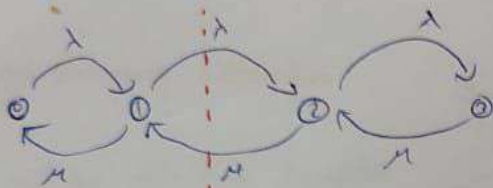
Using detailed balance eqⁿ,

$$\pi_{i+1} = \rho \pi_i$$

$$\sum \pi_i = 1$$

$$\pi_0 = 1 - \rho$$

$$= \rho^{i+1} \pi_0$$



detailed balance: left & right line mein right jaane ka ~~prob~~ ^{rate} or flow same as right to left

$$\pi_2(\lambda + \mu) = \pi_1 \cdot \lambda + \pi_3 \mu$$

$\pi Q = 0$ means it is balanced for 0, 1, 2, ...

$$\lambda \pi(n) + \cancel{\mu \pi(n)} = \lambda \pi(n-1) + \mu \pi(n+1)$$

equating these two doesn't make sense as it means that $\pi_n = \pi_{n+1}$

similar logic

gives us

$$\pi(n+1) = \frac{\lambda}{\mu} \pi_n = \left(\frac{\lambda}{\mu}\right)^n \pi_0$$

$$\rightarrow \sum \pi_i = 1$$

$$\Rightarrow \pi_0 (1 + \rho + \rho^2 + \dots) = 1$$

$$\Rightarrow \pi_0 = 1 - \rho$$

(assumed that $\lambda < \mu$, i.e., service is faster)

$$E[N] = \sum_n n \pi_n = \frac{\rho}{1 - \rho}$$